Roof Panel

Ford Ecat Electronic Parts Catalog

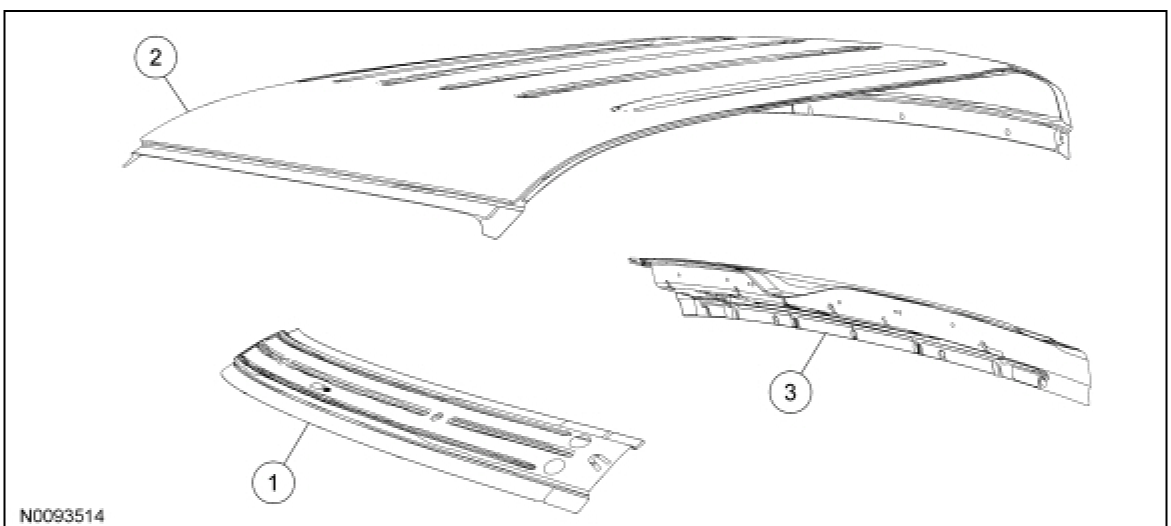
General Equipment

3 Phase Inverter Spot Welder 254-00002
Compuspot 700F Welder 190-50080
I4 Inverter Spot Welder 254-00014
Inverter Welder with MIG Welder 254-00015

Material

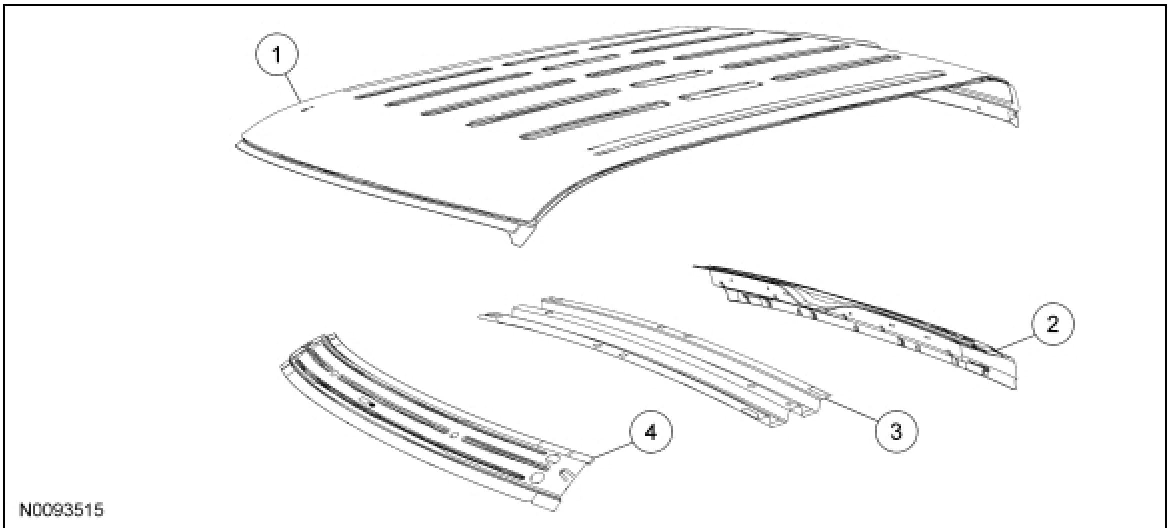
Item	Specification
Flexible Foam Repair Fusor® 121	—
Motorcraft® Metal Bonding Adhesive TA-1	—
Roof Ditch Sealer Fusor® 122EZ, 3M™ 08307	—
Motorcraft® Seam Sealer TA-2	—

Regular Cab



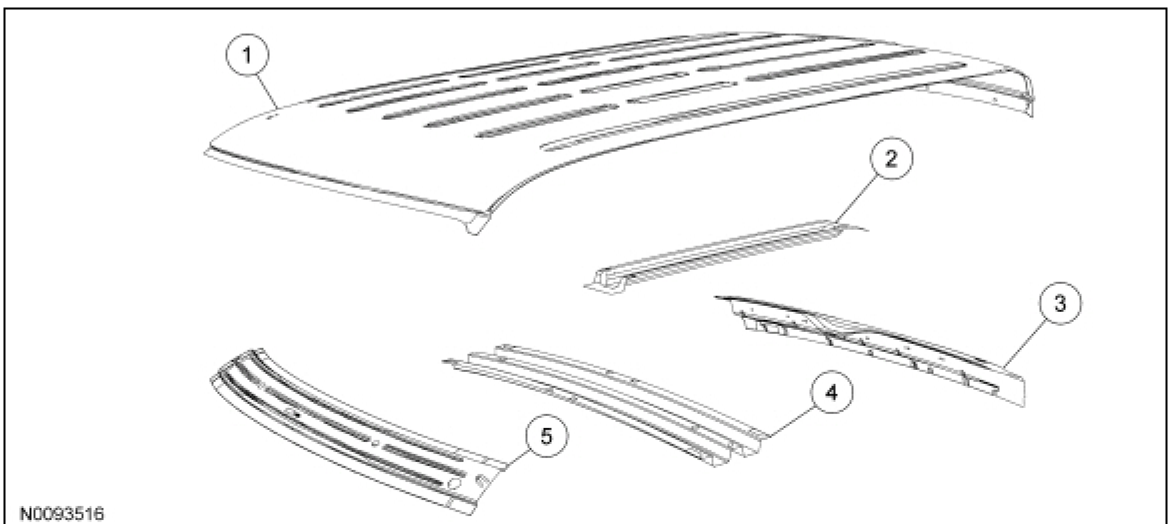
Item	Part Number	Description
1	0341803418	Windshield header — High-Strength Low Alloy (HSLA) 350 steel
2	5020250202	Roof panel — mild steel
3	4041840418	Back panel assembly — HSLA 350 steel

SuperCab



Item	Part Number	Description
1	5020250202	Roof panel — mild steel
2	4048440484	Back panel assembly — High-Strength Low Alloy (HSLA) 350 steel
3	502A74502A74	Roof bow — HSLA 350 steel
4	0341803418	Windshield header — HSLA 350 steel

SuperCrew Cab



Item	Part Number	Description
1	5020250202	Roof panel — mild steel
2	5029150291	Roof panel reinforcement — mild steel
3	4048440484	Back panel assembly — High-Strength Low Alloy (HSLA) 350 steel
4	502A74502A74	Roof bow — HSLA 350 steel

5	0341803418	Windshield header — HSLA 350 steel

Removal

WARNING: Collision damage repair must conform to the instructions contained in this workshop manual. Replacement components must be new, genuine Ford Motor Company parts. Recycled, salvaged, aftermarket or reconditioned parts (including body parts, wheels or safety restraint components) are not authorized by Ford.

Departure from the instructions provided in this manual, including alternate repair methods or the use of substitute components, risks compromising crash safety. Failure to follow these instructions may adversely affect structural integrity and crash safety performance, which could result in serious personal injury to vehicle occupants in a crash.

WARNING: Invisible ultraviolet and infrared rays emitted in welding can injure unprotected eyes and skin. Always use protection such as a welder's helmet with dark-colored filter lenses of the correct density. Electric welding will produce intense radiation, therefore, filter plate lenses of the deepest shade providing adequate visibility are recommended. It is strongly recommended that persons working in the weld area wear flash safety goggles. Also wear protective clothing. Failure to follow these instructions may result in serious personal injury.

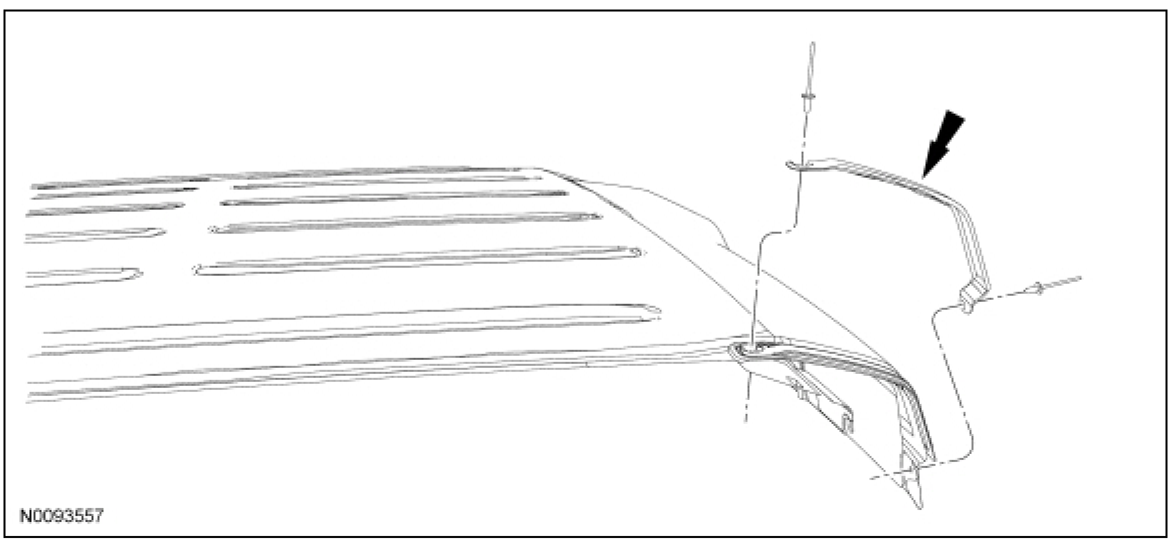
WARNING: Always wear protective equipment including eye protection with side shields, and a dust mask when sanding or grinding. Failure to follow these instructions may result in serious personal injury.

WARNING: Always refer to Material Safety Data Sheet (MSDS) when handling chemicals and wear protective equipment as directed. Examples may include but are not limited to respirators and chemically resistant gloves. Failure to follow these instructions may result in serious personal injury.

NOTICE: Be sure to adequately protect all glass, exterior finish and interior trim to avoid surface contamination from repair materials.

NOTE: *The driver and passenger sides of the roof panel are laser welded. Installation of a new roof panel will employ a weld-bonding technique. The use of mechanical welding and chemical bonding are both part of the installation technique.*

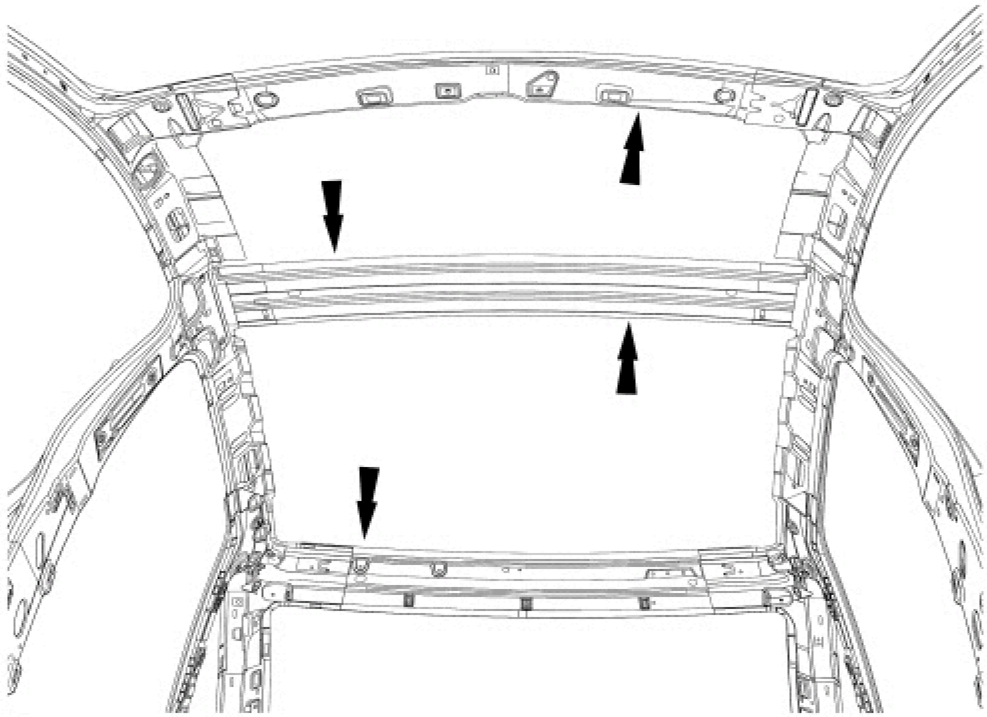
1. Remove the windshield and rear window glass. For additional information, refer to Section 501-05.
2. Remove the headliner. For additional information, refer to Section 501-05.
3. Remove the roof rear corner mouldings. The moulding is attached with rivets which must be drilled out to remove.



4. **NOTE:** *For installation purpose, particular attention must be paid to the depth (thickness) of the sealer removed.*

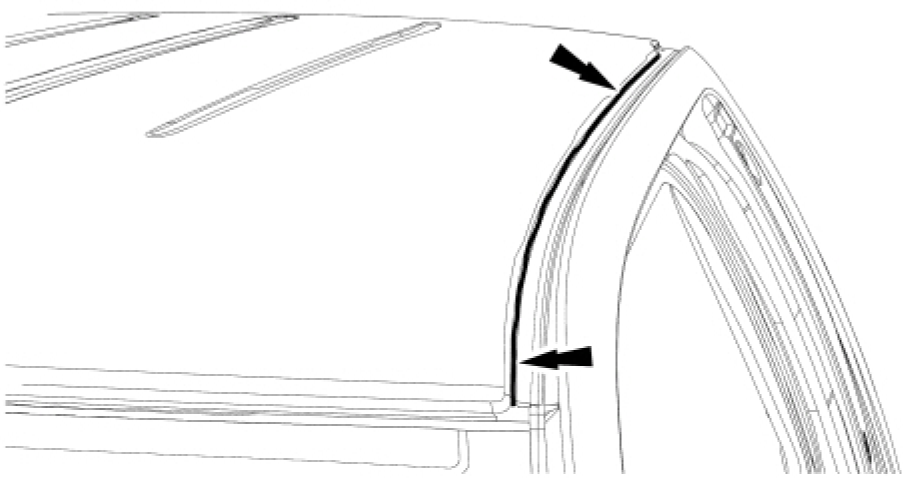
   Using a heat gun and a narrow scraper, remove the sealer along the entire length of the roof sides. Remove the sealer as completely as possible.

5. From inside the vehicle, leaving as much of the foam as possible intact, separate the NVH foam along the roof bows from the roof panel using a flexible and sharp broad scraper or knife.



N0039306

6. Remove (drill out) all the roof panel spot welds from the windshield header and back panel assembly.
7. Using a die grinder, carefully cut along the upper flange of the roof panel, the entire length of the panel.



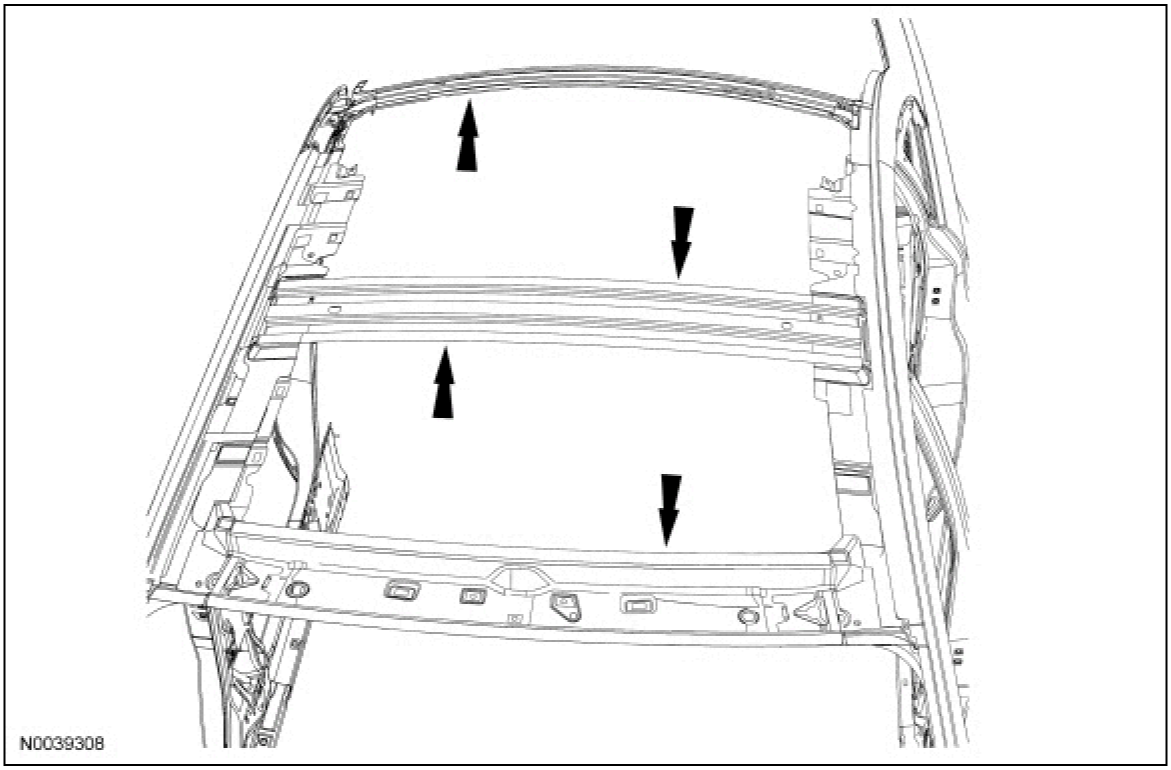
N0093556

8. With the help of an assistant, remove the roof panel from the vehicle.
9. Using an air chisel equipped with a flat chisel or similar, remove the remaining portion of the roof panel from the roof ditch area.

Installation

1. Using a hammer and dolly, straighten any damage caused to flange areas during removal.
2. Using a grinder, carefully prepare the roof ditch area for the new roof panel.
3. With the help of an assistant, position the new roof panel on the vehicle.
4. Correctly align and index-mark the positioning of the panel-to-vehicle and remove the panel.

5. Apply a 13 mm (0.51 in) tall bead of seam sealer directly to the NVH foam remaining on the roof bows. Apply a bead of metal bonding adhesive following label instructions, along the roof ditch area on each side.



6. Reinstall the roof panel aligning it to the index marks made during the test fitting. If adjustment is necessary, do not lift the roof panel, slide to reposition. Make sure good contact is made with the sealer and adhesive and carefully clamp the roof panel on all sides.

7. **NOTE:** *The roof panel is not welded along the roof ditch flange, only metal bonding adhesive is used.*

   Using a resistance spot welder, weld the roof panel to the vehicle in the windshield and back panel assembly areas. Spot welds should match factory welds in both location and quantity. Do not place a new spot weld directly over an original weld location. Plug weld hole should equal 8 mm (0.31 in) diameter.

8. Using a Metal Inert Gas (MIG) welder, seam weld the roof panel at each corner with 10 mm (0.39 in) long weld beads.
   - Apply flexible foam as necessary between roof panel and reinforcements to maintain NVH performance.

9. Prepare the repair area for refinish material application following Ford-approved paint recommendations.

10. Apply a quality primer-sealer following the paint manufacturer's recommendation for both compatibility and application.

11. **NOTE:** *Always use refinishing materials from a single paint manufacturer. Combining products from more than one manufacturer may result in refinishing material incompatibility issues and potential void of warranty for each product applied.*

    If required, follow up with a quality primer-surfacer following Ford-approved paint recommendations. Block sand to level using the recommended grit sandpaper.

12. **NOTICE: Failure to purge air bubbles from the roof ditch sealer may result in air trapped in the roof ditch which may cause refinishing materials to pop.**

    Stand the roof ditch sealer vertically (nose up) for a minimum of 30 minutes prior to application to permit trapped air to rise to the top so it may be purged.

13. Position the vehicle so it is level front-to-rear.
    - Using masking tape, form a dam at each end of the roof ditch (both sides) to prevent sealer run-off.

14. Apply the roof ditch sealer following product directions to a depth equal to the sealer noted during removal.
15. Allow the sealer to cure 40-60 minutes at room temperature before refinishing. If roof ditch sealer has been allowed to cure in excess of 2 hours, scuff the surface using a red scuff pad.
16. Refinish the roof and repair area following Ford-approved paint recommendations.
17. Allow refinishing materials sufficient cure time to permit safe handling (refer to paint manufacturer's specification) and install the roof corner mouldings.
18. Install the headliner. For additional information, refer to Section 501-05.
19. Install the windshield and rear window glass. For additional information, refer to Section 501-11.